DESCRIPTION OF COURSEWORK

Course Code	EEE303
Course Name	Electrical Machinery & Drives
Lecturer	Dr. Ng Zi Neng
Academic Session	2023/09
Assessment Title	Assignment – Performance evaluation of a Synchronous Motor

A. Introduction/ Situation/ Background Information

This assignment is designed to assess student's knowledge and understanding on synchronous motor. A synchronous motor is an AC electric motor in which, at steady state, the rotation of the shaft is synchronized with the frequency of the supply current. As it is not self-starting, an external force is needed to accelerate the rotor during the start. With damper winding, synchronous motor is able to start as an induction motor without the excitation DC current to the field winding. When the rotor reaches near synchronous speed, the DC source is applied to the rotor's field winding to produce constant magnetic flux, and lock in magnetically with the stator's rotating magnetic field generated by the three-phase AC supply.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 2 Investigate the expected performance of DC machines, synchronous machines and induction motor.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating

published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

The students are expected to work independently as a group to gather information from online literatures and simulate a three-phase AC synchronous motor using **MATLAB Simulink**.

1. To setup AC synchronous motor block diagram.
2. To design a block diagram to start and run the AC synchronous motor to its rated speed.
3. To evaluate the performance of the AC synchronous motor based on:
   a. The variation of mechanical load on the load angle, stator current, electromagnetic torque, and efficiency.
   b. The variation of field voltage on the load angle, stator current, electromagnetic torque, and efficiency.
4. Identify the pull-in and pull-out torque.

The work will be presented in the form of a written report (one report per group).

Proper formatting and style are expected for the report:

- Use the Assignment Cover Page and fill in each group member's name and other details.
- The Times New Roman typeface at font size 12 and 1.5 line spacing should be used in the main text throughout the report. Bold for the main topic and sub-topic with proper numberings.

- All paragraphs in a subsection should align with both margin (Justify text).
- Figures and tables have to be captioned and placed according to the standard format, i.e., Figure 1.1 “caption” is placed after the figure and Table 1.1 “caption” is placed before the table.
- Cite references with APA citation format.
- Naming convention for pdf file: EEE303Assignment_ Group1.pdf

The submission due date: **Week 14, Friday 22/12/2023 23:55 Malaysian time**.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Introduction	10
2.	Methodology and simulation setup	30
3.	Results and discussions	40
4.	Conclusions and references	15
5.	Report formatting	5
	TOTAL COURSEWORK MARK	**100**

F. Tasks

The contents of your report should comprehensively cover the following subject:

1. Introduction
    - Operating principle and key parameters/nameplate of a synchronous motor
2. Methodology and simulation setup
    - Block diagram of a complete three-phase AC synchronous motor setup (Synchronous Machine block can be obtained in Simscape → Electrical → Specialized Power Systems → Fundamental Blocks → Machines → **Synchronous Machine SI Fundamental**, as shown in Figure 1).
    - Double click the Synchronous Machine SI Fundamental block, select **Preset model 05: 50 Hz 400 V 60 kVA 1500 RPM**.
    - Include other blocks such as voltage supply, scopes, gain, etc. Identify the function of each block and explain the key values set in all blocks’ parameters (by double clicking the block).

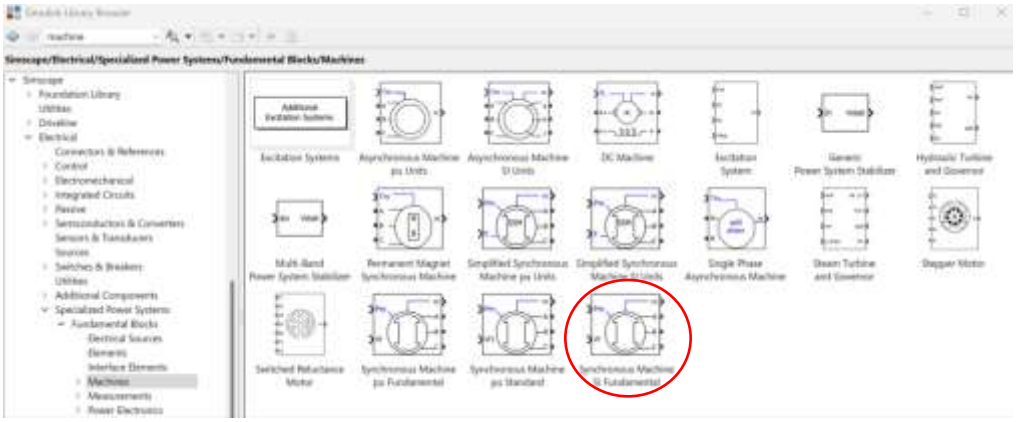


Figure 1. Synchronous Machine SI Fundamental block

3. Results and discussions
    - Show the parameter (voltage/current/speed/torque) waveforms as the motor starts from stationary to the rated speed.
    - Elucidate the motor's behavior due to the variation of mechanical loads, show the relevant waveforms.
    - Elucidate the motor's behavior due to the variation of field voltage, show the relevant waveforms.
    - Obtain the steady-state values once the speed is stabilized, usually after the motor runs for 5 seconds or more. Plot relevant graphs i.e., load angle vs. mechanical load, load angle vs. field voltage, etc. For alternating current and voltage, measure the root mean square (RMS) value upon reaching steady state.
    - Analyze the results and identify the correlations of the variables to the load angle, stator current, electromagnetic torque, etc. Support results with relevant theories and equations.
4. Conclusions and references
    - Summary of key results
    - Citation of all sources used in the report with APA style referencing
5. Report formatting
    - Consistent font sizes for main/sub-topics and text
    - Include all required sections and sub-sections
    - Table and figure captions must be numbered and mentioned in text

APPENDIX 1

Component Title	Assignment				Percentage (%)	100	
Criteria	Score and Descriptors					Weight (%)	Marks
	Excellent (5)	Good (4)	Average (3)	Need Improvement (2)	Poor (1)		
Introduction	Excellently written, concise introduction with important information that are relevant to the topic.	Well written, concise introduction with important information that are relevant to the topic.	Averagely written, concise methodology, consists some irrelevant information.	Averagely written, lengthy methodology, consists many irrelevant information.	Poorly written, lengthy methodology, consists many irrelevant information.	10	
Methodology and simulation setup	Excellently written, concise methodology with important information that are relevant to the topic.	Well written, concise methodology with important information that are relevant to the topic.	Averagely written, concise methodology, consists some irrelevant information.	Averagely written, lengthy methodology, consists many irrelevant information.	Poorly written, lengthy methodology, consists many irrelevant information.	30	
Results and discussions	Completed all experiments and data collection. All measured data, calculations, and graphs/plots are correct. Able to evaluate the results with prudent judgment. Analysis is done with excellent description of the relationship involved and explanation with relevant theories.	Completed all experiments and data collection. Most measured data, calculations, and graphs/plots are correct. Results are compared and analyzed with good description of the relationship involved and with relevant theories.	Completed most experiments and data collection. Some measured data, calculations, and graphs/plots are incorrect. Results analysis are done with adequate description of the relationship involved.	Completed most experiments, but many measured data, calculations, and graphs/plots are incorrect. Only restatement of the results without commenting on the expected key points.	Experiments not completed. Most measured data, calculations, and graphs/plots are incorrect. Discussion shows little understanding of what the experiment is all about.	40	
Conclusions and references	The aim of the experiment has been achieved, the key features of the methods used and most important results are summarized.	The aim of the experiment has been achieved, most key features of the methods used and results are summarized.	A conclusion is drawn, but only supported by some experimental evidence.	A conclusion is drawn, but not supported by any experimental evidence.	No sensible conclusion. No comparisons and evidence.	15	
Report formatting	All sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and mentioned in the main text. All figures are clear with important labels and comments. Title and the axis of all plots are well labelled, clear, and easy to read.	Most sections are presented in a neat and tidy manner with consistent format and style. Complete captions for tables and figures are given and most are mentioned in the main text. Most figures are clear with labels and comments. Title and the axis of most plots are well labelled and clear.	Some sections are inconsistent in terms of format and style. Most captions for tables and figures are given and mentioned in the main text. Some figures and components in the circuit diagram are not clear and missing important labels and comments. Some title and axis of plots are not well labelled and clear.	Most sections are inconsistent in terms of format and style. Incomplete captions for tables and figures and most are not mentioned in the main text. Most figures and components in the circuit diagram are not clear and missing important labels and comments. Most title and axis of plots are not well labelled and clear.	Messy report and difficult to read with inconsistent format and style. No captions for tables and figures provided. Blurry figures and plots with missing labels provided. Figures and plots are difficult to read and understand.	5	
					TOTAL	**100**	

Note to students: Please attach this appendix together with the submission of coursework.